

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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SCIENCE

1113/02

Paper 2

October 2020

45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

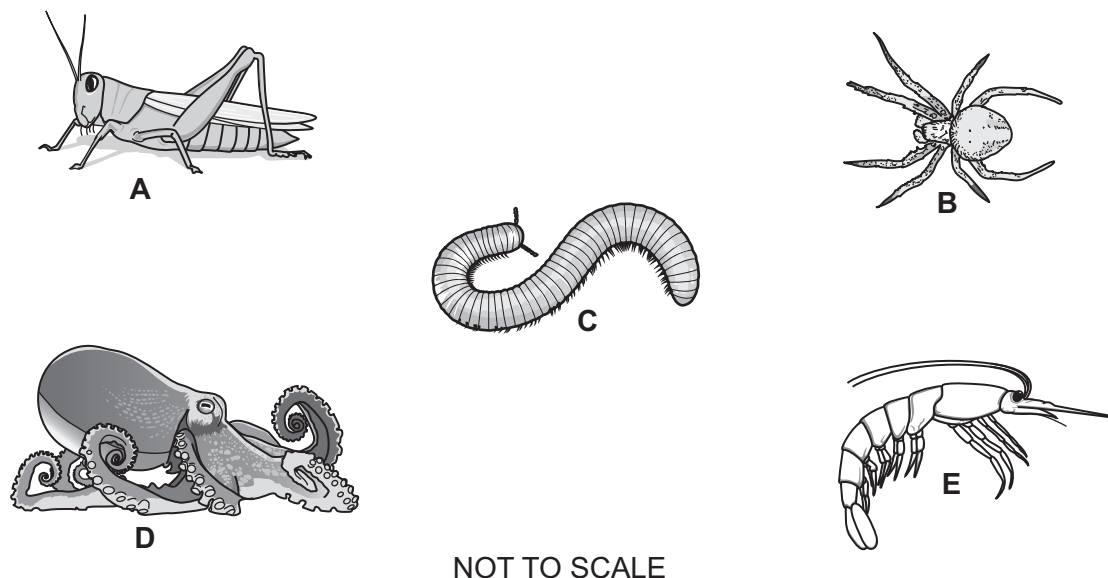
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Blank pages are indicated.

1 The diagrams **A**, **B**, **C**, **D** and **E** show five different invertebrate animals.



(a) (i) Which one of the animals is an insect?

Choose from **A**, **B**, **C**, **D** or **E**.

.....

Give a reason for your answer.

..... [1]

(ii) Which one of the animals is an arachnid?

Choose from **A**, **B**, **C**, **D** or **E**.

.....

Give a reason for your answer.

..... [1]

(b) Which one of the animals is **not** an arthropod?

Choose from **A**, **B**, **C**, **D** or **E**.

.....

Give a reason for your answer.

..... [1]

2 This question is about the three states of matter.

(a) A gas is blown into a balloon. The balloon changes shape.

Why does the balloon change shape?

Tick (✓) the box next to the **correct** answer.

The particles of the gas expand to fill the space.

☐

The particles of the gas get bigger.

☐

The particles of the gas hit the surface of the balloon more often.

☐

The particles of the gas slow down.

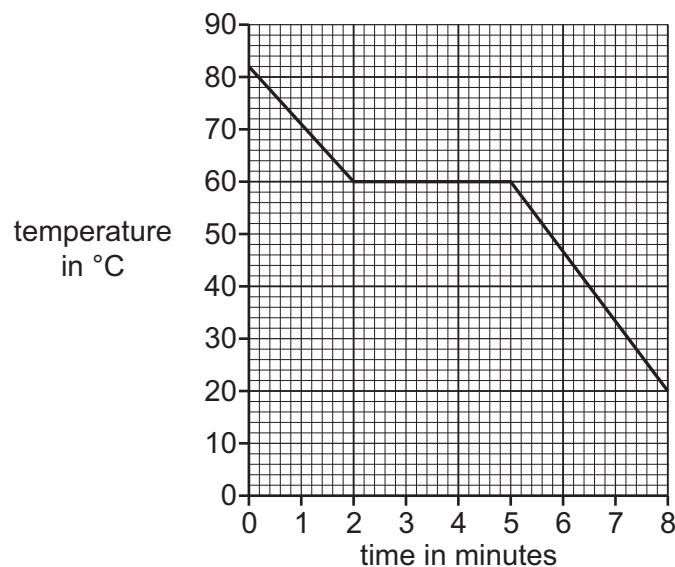
☐

[1]

(b) Rajiv investigates the cooling curve of a substance.

He measures the temperature of a hot liquid every minute.

The graph shows his results.



(i) What is the temperature of the hot liquid at the start?

..... °C

[1]

(ii) Name the process that happens when a liquid changes into a solid.

.....

[1]

(iii) At what temperature does the liquid change into a solid?

..... °C

[1]

3 Complete the sentences about thermal (heat) energy transfer.

Choose words from the list.

conduction

convection

radiation

Thermal (heat) energy can be transferred from one place to another place.

When particles are involved, the processes are and

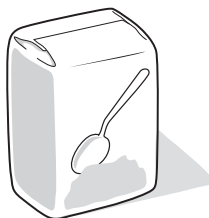
When electromagnetic waves are involved, the process is

[2]

4 Look at the list of different energy sources.



wood



sugar



coal



**crude oil
(petroleum)**



**propane
(LPG)**



sunflower oil

Which three of these energy sources are **non-renewable**?

- 1
- 2
- 3

[1]

- 5 Complete the sentences about changes that happen in the human body during adolescence.

Choose words or phrases from the list.

at the same time as body hair breasts earlier than
enzymes hormones later than proteins

During adolescence, the human body begins to change.

Both sexes begin to grow

This is due to the increased amount of made by the sex organs.

On average, females become sexually mature males.

[3]

- 6 Chen investigates some reactions.

He measures the temperature of the reactants at the start of the reaction.

He measures the temperature of the products at the end of the reaction.

Look at his results.

reaction	temperature at start in °C	temperature at end in °C	temperature change in °C	type of reaction
A	20	15	-5	endothermic
B	20	30		
C	15	30		
D	25	15		

- (a) Complete the table.

[2]

- (b) Which reaction has the **greatest** energy change?

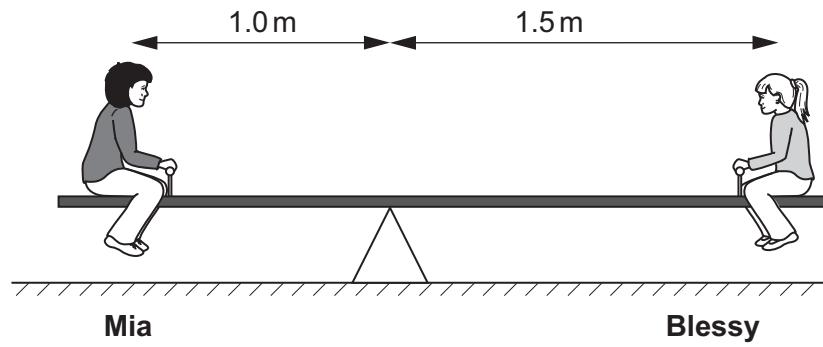
.....

Explain how you know.

.....

[2]

- 7 Mia and Blessy sit on a balanced seesaw.



Blessy has a weight of 400 N.

Calculate the weight of Mia.

weight of Mia N [2]

8 Some plants live in dry places where there is very little rainfall.

(a) These plants often have a large network of roots.

Describe **two** different functions of roots.

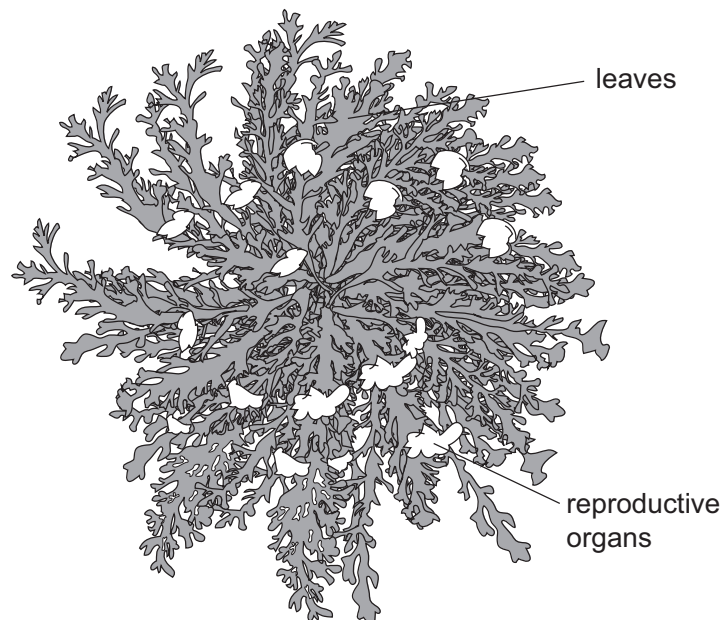
- 1
- 2

[2]

(b) When it is dry, resurrection plants are rolled up.

In wet conditions, resurrection plants unroll and flatten out.

This exposes the plant's leaves and reproductive organs.



Explain why these changes are important to the resurrection plant's survival.

(i) Unrolling to expose its leaves.

.....

.....

..... [2]

(ii) Unrolling to expose its reproductive organs.

.....

..... [2]

9 Lily investigates the boiling points of some liquids.

(a) She uses a Bunsen burner to heat 20 cm^3 of each liquid in a beaker.

Which equipment should Lily use to measure the boiling point of the liquid?

..... [1]

(b) Write down **one** safety precaution that Lily should take.

..... [1]

10 Mike explains that different materials have different densities.

He says,

‘Materials that are less dense than water will float.

Water has a density of 1 g/cm^3 .’

He finds out the densities of different materials.

material	density in g/cm^3
gold	19.3
plastic	1.05
pumice stone	0.251
silver	10.5
wood	0.715

(a) Predict which **two** materials will float.

Choose from the table.

..... and [1]

(b) Mike collects some water from the sea.

He tries to float the materials from the table in the seawater.

Now **three** of these materials float.

Use information from the table to explain why.

.....

 [2]

11 The seeds of cereal plants provide food for humans.

The diagram shows five varieties of a cereal plant.

A 	B 	C 
produces lots of seeds	small leaves	does not need much water
D 	E 	
long stems	shallow roots	

Scientists want to produce a new variety of this cereal plant.

This cereal plant must:

- provide lots of food
- survive in dry conditions.

(a) Which **two** plants should the scientists use to produce the new variety of cereal plant?

Choose from **A**, **B**, **C**, **D**, and **E**.

..... and [1]

(b) Describe what the scientists do to produce the new variety.

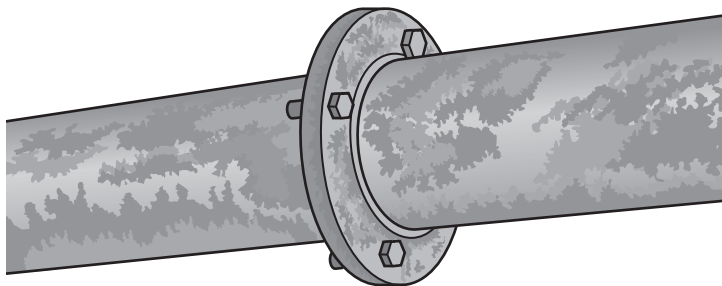
.....

 [2]

(c) Name the process scientists use to produce new varieties of plants and animals.

..... [1]

12 Look at the picture of an iron pipe.



The iron has reacted with water and a gas in the air to form hydrated iron oxide.

(a) What word describes this reaction?

..... [1]

(b) Which gas in the air reacts with the iron?

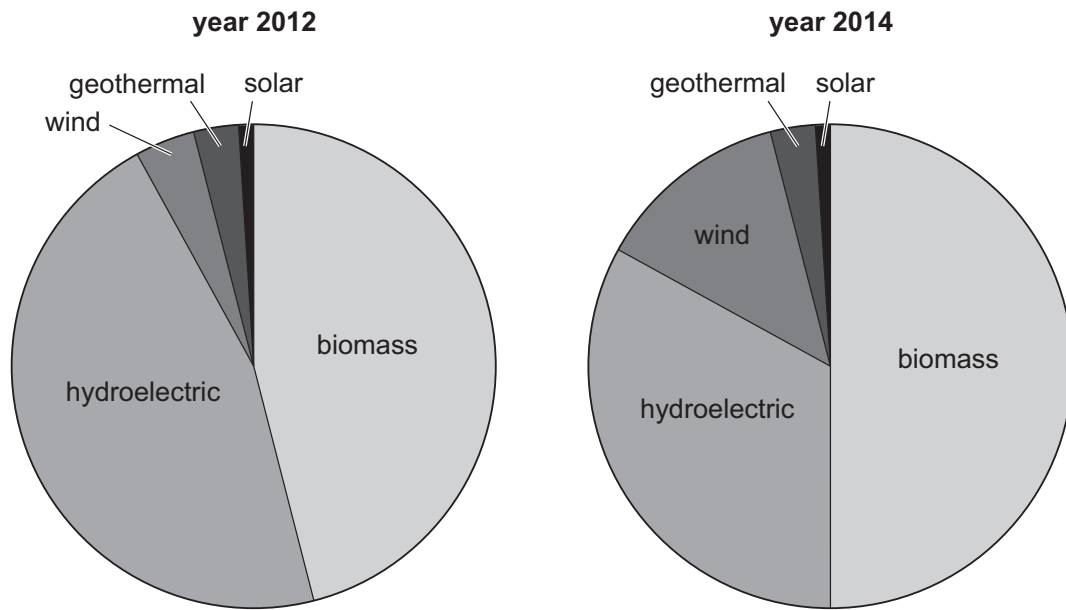
..... [1]

(c) This reaction is **not** useful.

Explain why.

..... [1]

- 13 Safia finds information about renewable energy resources used in a country for the years 2012 and 2014.



- (a) Estimate the percentage of renewable energy provided by biomass in 2012.

..... %

[1]

- (b) The percentage of the renewable energy resources used changed from 2012 to 2014.

Which percentage increased the most?

Circle the correct answer.

biomass
geothermal
hydroelectric
solar
wind

[1]

- (c) The energy needs of the world are increasing.

Why is it important to develop **renewable** energy resources?

.....
..... [1]

14 Look at the diagram.

It shows part of the Periodic Table.

		H						He		
Li	Be				B	C	N	O	F	Ne
Na	Mg				Al	Si	P	S	Cl	Ar
K	Ca	transition elements								

(a) Write down the chemical symbol of an element in the same **group** as chlorine.

.....

[1]

(b) Write down the chemical symbol of an element in the same **period** as sodium.

.....

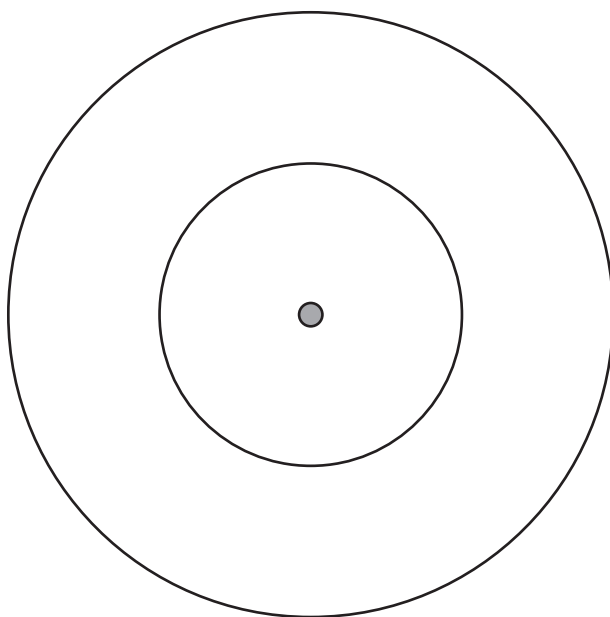
[1]

(c) Write down the chemical symbol of the element with an atom with only 8 protons.

.....

[1]

(d) Complete the diagram to show the electronic structure of lithium, Li.



[1]

15 The boxes show some ideas about the Solar System.

(a) Match each **idea** to the correct **scientist**.

Draw only **two** lines.

ideas

scientist

The first scientist to suggest that the planets orbited the Sun.

Galileo

Rutherford

Copernicus

Built a telescope and looked at Jupiter and its moons. This showed that everything does not rotate around the Earth.

Pasteur

Darwin

[2]

(b) Here is a list of bodies found in the Solar System.

Earth

Mars

Moon

Sun

Which one of these bodies is visible because it **emits** light?

..... [1]

(c) Which **two** of these bodies are visible because they **reflect** light?

Mars

Moon

North Star

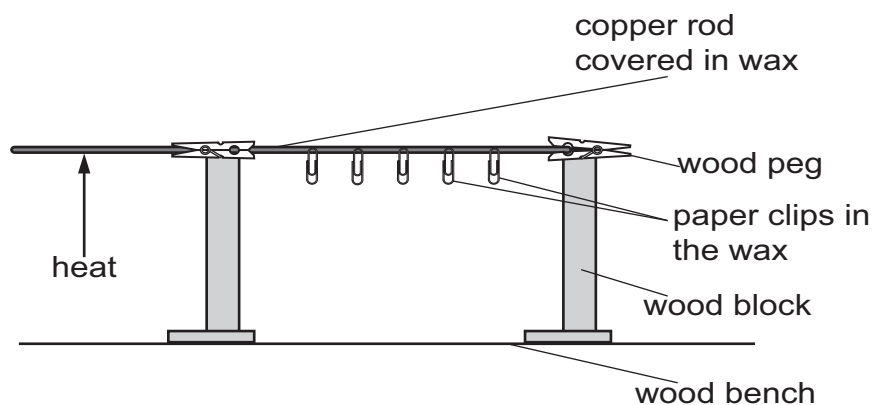
Sun

..... and [1]

16 Aiko and Oliver investigate thermal (heat) energy.

They heat a copper rod covered in wax.

Here is the apparatus they use.



When the copper rod is hot, the wax melts and the paper clips fall onto the bench.

(a) Aiko removes the copper rod. Oliver tells her to be careful.

Complete the sentences.

Aiko must be careful because

When she removes the copper rod, she uses

[1]

(b) Here are the results.

distance of paper clip from heat in cm	time for paper clip to fall in seconds
6	3.2
7	4.3
8	5.4
9	6.6
10	7.9

Aiko says it is a good idea to repeat the experiment.

Explain why this is a good idea.

.....

[1]

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